Determinant: Questions and Full Solutions

Question 01

English

Let b and c be real numbers and let $P : \mathbb{R} \to \mathbb{R}$ be defined by $P(x) = 3x^2 + bx + c$. Then

$$\begin{vmatrix} P(0) & P(1) & P(2) \\ P(1) & P(2) & P(3) \\ P(2) & P(3) & P(4) \end{vmatrix}$$

is equal to

(A) 0
(B) 216
(C) -216
(D) -108

Only one option is correct.

हिन्दी

माना कि b तथा c वास्तविक संख्याएँ हैं और $P : \mathbb{R} \to \mathbb{R}$, $P(x) = 3x^2 + bx + c$ द्वारा परिभाषित है। तब

$$\begin{vmatrix} P(0) & P(1) & P(2) \\ P(1) & P(2) & P(3) \\ P(2) & P(3) & P(4) \end{vmatrix}$$

बराबर है

(A) 0
(B) 216
(C) -216
(D) -108

केवल एक विकल्प सही है।

Solution

Write $u_n = P(n)$. For $P(x) = 3x^2 + bx + c$: $u_{n+1} - u_n = 6n + 3 + b$ and $u_{n+2} - 2u_{n+1} + u_n = 6$ for every n.

Apply $R_3 \to R_3 - 2R_2 + R_1$ (value unchanged, since a multiple of other rows is added). The new third row is $(u_2 - 2u_1 + u_0,\ u_3 - 2u_2 + u_1,\ u_4 - 2u_3 + u_2) = (6, 6, 6)$.

Apply $R_2 \to R_2 - R_1$. The new second row is $(u_1 - u_0,\ u_2 - u_1,\ u_3 - u_2) = (3 + b,\ 9 + b,\ 15 + b)$. The first row stays $(c,\ 3 + b + c,\ 12 + 2b + c)$.

Apply $C_2 \to C_2 - C_1$: the second column becomes $(3 + b,\ 6,\ 0)$.

Apply $C_3 \to C_3 - 2C_2 + C_1$ (using the original second column): first row $12+2b+c-2(3+b+c)+c = 6$, second row $15 + b - 2(9 + b) + (3 + b) = 0$, third row $6 - 12 + 6 = 0$.

$$\begin{vmatrix} c & 3 + b & 6 \\ 3 + b & 6 & 0 \\ 6 & 0 & 0 \end{vmatrix}$$

Expanding along the third column: $6 \begin{vmatrix} 3 + b & 6 \\ 6 & 0 \end{vmatrix} = 6(0 - 36) = -216$.

Why b and c do not matter: if P were linear, $u_{n+2} - 2u_{n+1} + u_n = 0$, so $R_3 = 2R_2 - R_1$ and the determinant is 0; only the x^2 part survives, and it gives $-8 \cdot 3^3 = -216$.

Option (D) is the value $-4 \cdot 27$ obtained by forgetting the factor 2 in the second difference.

Correct Option: (C)

Question 02

English

Let $f : \mathbb{R} \to \mathbb{R}$ be defined by

$$f(x) = \begin{vmatrix} 1 & 1 & 1 \\ \cos x & \cos 2x & \cos 3x \\ \cos 2x & \cos 4x & \cos 6x \end{vmatrix}.$$

Then the number of distinct $x \in [0, \pi]$ for which $f(x) = 0$ is

(A) 6
(B) 5
(C) 7
(D) 4

Only one option is correct.

हिन्दी

माना कि $f : \mathbb{R} \to \mathbb{R}$

$$f(x) = \begin{vmatrix} 1 & 1 & 1 \\ \cos x & \cos 2x & \cos 3x \\ \cos 2x & \cos 4x & \cos 6x \end{vmatrix}$$

द्वारा परिभाषित है। तब $[0, \pi]$ में उन भिन्न x की संख्या, जिनके लिए $f(x) = 0$ है,

(A) 6
(B) 5
(C) 7
(D) 4

केवल एक विकल्प सही है।

Solution

Since $\cos 2\theta = 2\cos^2\theta - 1$, the third row is $2(\cos^2 x, \cos^2 2x, \cos^2 3x) - (1, 1, 1)$, a linear combination of the first row and the row of squares.

Apply $R_3 \to R_3 + R_1$ (adding a multiple of a row does not change the determinant). The third row becomes $(2\cos^2 x,\ 2\cos^2 2x,\ 2\cos^2 3x)$; take the factor 2 out.

With $a = \cos x$, $b = \cos 2x$, $c = \cos 3x$ the remaining determinant is the Vandermonde determinant $\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix} = (b-a)(c-a)(c-b)$.

So $f(x) = 2(\cos 2x - \cos x)(\cos 3x - \cos x)(\cos 3x - \cos 2x)$, and $f(x) = 0$ exactly when two of $\cos x, \cos 2x, \cos 3x$ are equal.

$\cos 2x = \cos x$: $2x = 2k\pi \pm x$, so $x = 2k\pi$ or $x = \frac{2k\pi}{3}$. In $[0, \pi]$: $x = 0,\ \frac{2\pi}{3}$.

$\cos 3x = \cos x$: $3x = 2k\pi \pm x$, so $x = k\pi$ or $x = \frac{k\pi}{2}$. In $[0, \pi]$: $x = 0,\ \frac{\pi}{2},\ \pi$.

$\cos 3x = \cos 2x$: $3x = 2k\pi \pm 2x$, so $x = 2k\pi$ or $x = \frac{2k\pi}{5}$. In $[0, \pi]$: $x = 0,\ \frac{2\pi}{5},\ \frac{4\pi}{5}$.
The value 0 occurs in all three lists and must be counted once. The distinct zeros are $0,\ \frac{2\pi}{5},\ \frac{\pi}{2},\ \frac{2\pi}{3},\ \frac{4\pi}{5},\ \pi$. Hence there are 6 zeros. Counting 0 more than once gives 7 or more; missing the end point π gives 5.

Correct Option: (A)

Question 03

English

For each positive integer n, let

$$D_n = \begin{vmatrix} n & n+1 & n+2 \\ n+2 & n & n+1 \\ n+1 & n+2 & n \end{vmatrix}.$$

Then the number of integers n with $1 \le n \le 100$ for which D_n is the square of an integer is

(A) 10
(B) 9
(C) 3
(D) 8

Only one option is correct.

हिन्दी

प्रत्येक धनात्मक पूर्णांक n के लिए माना कि

$$D_n = \begin{vmatrix} n & n+1 & n+2 \\ n+2 & n & n+1 \\ n+1 & n+2 & n \end{vmatrix}$$

है। तब $1 \le n \le 100$ वाले उन पूर्णांकों n की संख्या, जिनके लिए D_n किसी पूर्णांक का वर्ग है,

(A) 10
(B) 9
(C) 3
(D) 8

केवल एक विकल्प सही है।

Solution

The matrix is circulant: each row is the previous one shifted cyclically, so every row and every column has sum $3n + 3$.

Apply $C_1 \to C_1 + C_2 + C_3$; the first column becomes $(3n+3, 3n+3, 3n+3)$, and $3(n+1)$ comes out as a common factor.

$$D_n = 3(n+1)\begin{vmatrix} 1 & n+1 & n+2 \\ 1 & n & n+1 \\ 1 & n+2 & n \end{vmatrix}.$$

Apply $R_2 \to R_2 - R_1$ and $R_3 \to R_3 - R_1$: $\begin{vmatrix} 1 & n+1 & n+2 \\ 0 & -1 & -1 \\ 0 & 1 & -2 \end{vmatrix} = (-1)(-2) - (-1)(1) = 3.$

So $D_n = 9(n+1)$. This agrees with the circulant formula $a^3 + b^3 + c^3 - 3abc = (a+b+c) \cdot \frac{1}{2}[(a -$

$b)^2 + (b-c)^2 + (c-a)^2] = (3n+3) \cdot 3$.

Since 9 is a square, D_n is a perfect square exactly when $n+1$ is a perfect square.

For $1 \le n \le 100$ we have $2 \le n+1 \le 101$, and the squares in this range are $4, 9, \ldots, 100$, that is k^2 for $k = 2, \ldots, 10$.

Hence 9 values of n (namely $n = 3, 8, 15, \ldots, 99$). Including $n+1 = 1$ (outside the range) would give 10.

Correct Option: (B)

Question 04

English

Let ω be a complex cube root of unity with $\omega \neq 1$. The equation

$$\begin{vmatrix} x & 1 & \omega \\ \omega & x & 1 \\ 1 & \omega & x \end{vmatrix} = 0$$

in the complex variable x has a repeated root α and another root β. Then $\dfrac{\beta}{\alpha}$ is equal to

(A) $-\dfrac{1}{2}$
(B) 2
(C) ω
(D) -2

Only one option is correct.

हिन्दी

माना कि ω इकाई का एक सम्मिश्र घनमूल है तथा $\omega \neq 1$ है। सम्मिश्र चर x में समीकरण

$$\begin{vmatrix} x & 1 & \omega \\ \omega & x & 1 \\ 1 & \omega & x \end{vmatrix} = 0$$

का एक पुनरावृत्त मूल α तथा एक अन्य मूल β है। तब $\dfrac{\beta}{\alpha}$ बराबर है

(A) $-\dfrac{1}{2}$
(B) 2
(C) ω
(D) -2

केवल एक विकल्प सही है।

Solution

Each row is the previous row shifted one place to the right, so the determinant is circulant with first row $(a, b, c) = (x, 1, \omega)$.

For a circulant, $\Delta = a^3+b^3+c^3-3abc = (a+b+c)(a+b\omega+c\omega^2)(a+b\omega^2+c\omega)$, because $1+\omega+\omega^2 = 0$ and $\omega^3 = 1$.

Direct expansion: $\Delta = x^3 + 1 + \omega^3 - 3x\omega = x^3 - 3\omega x + 2$.

First factor: $x + 1 + \omega = x - \omega^2$.

Second factor: $x + 1 \cdot \omega + \omega \cdot \omega^2 = x + \omega + 1 = x - \omega^2$.

Third factor: $x + 1 \cdot \omega^2 + \omega \cdot \omega = x + 2\omega^2$.

So $\Delta = (x - \omega^2)^2(x + 2\omega^2)$: two of the three linear factors coincide because $\omega^3 = 1$ makes $c\,\omega^2 = 1$.

Check: the product of the roots is $\omega^4 \cdot (-2\omega^2) = -2\omega^6 = -2$, which equals -2 from $x^3 - 3\omega x + 2$; the sum of the roots is $2\omega^2 - 2\omega^2 = 0$, matching the missing x^2 term.

The repeated root is $\alpha = \omega^2$ and the other root is $\beta = -2\omega^2$, so $\dfrac{\beta}{\alpha} = -2$, whichever non-real cube root ω is.

Correct Option: (D)

Question 05

English

For a real number k, let $p, q, r : \mathbb{R} \to \mathbb{R}$ be defined by $p(x) = x^2 + 1$, $q(x) = x^2 + x$ and $r(x) = 2x^2 + kx + k^2$. Then the sum of all real values of k for which

$$\begin{vmatrix} p(1) & p(2) & p(3) \\ q(1) & q(2) & q(3) \\ r(1) & r(2) & r(3) \end{vmatrix} = 0$$

is

(A) -1

(B) 1

(C) -2

(D) 3

Only one option is correct.

हिन्दी

वास्तविक संख्या k के लिए माना कि $p, q, r : \mathbb{R} \to \mathbb{R}$, $p(x) = x^2 + 1$, $q(x) = x^2 + x$ तथा $r(x) = 2x^2 + kx + k^2$ द्वारा परिभाषित हैं। तब k के उन सभी वास्तविक मानों का योग, जिनके लिए

$$\begin{vmatrix} p(1) & p(2) & p(3) \\ q(1) & q(2) & q(3) \\ r(1) & r(2) & r(3) \end{vmatrix} = 0$$

है,

(A) -1

(B) 1

(C) -2

(D) 3

केवल एक विकल्प सही है।

Solution

For a polynomial $s(x) = ax^2 + bx + c$ the row $(s(1), s(2), s(3))$ equals $a(1, 4, 9) + b(1, 2, 3) + c(1, 1, 1)$.

Hence the determinant equals $\begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 2 & k & k^2 \end{vmatrix} \cdot \begin{vmatrix} 1 & 4 & 9 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix}$ (rows of coefficients times the rows $(j^2), (j), (1)$).

$$\begin{vmatrix}1 & 4 & 9\\ 1 & 2 & 3\\ 1 & 1 & 1\end{vmatrix} = 1(2-3) - 4(1-3) + 9(1-2) = -2 \neq 0.$$

$$\begin{vmatrix}1 & 0 & 1\\ 1 & 1 & 0\\ 2 & k & k^2\end{vmatrix} = 1(k^2 - 0) - 0 + 1(k-2) = k^2 + k - 2.$$

So the determinant is $-2(k^2 + k - 2)$. It vanishes exactly when the coefficient rows are dependent, that is when r is a linear combination $\lambda p + \mu q$: $\mu = k$ (x-coefficient), $\lambda = k^2$ (constant), $\lambda + \mu = 2$ (x^2-coefficient).

$k^2 + k - 2 = (k-1)(k+2) = 0$, so $k = 1$ ($r = p + q$) or $k = -2$ ($r = 4p - 2q$).

Check for $k = 1$: rows $(2, 5, 10)$, $(2, 6, 12)$, $(4, 11, 22)$, and the third is the sum of the first two.

The sum of the values is $1 + (-2) = -1$. The product -2 is option (C).

Correct Option: (A)

Question 06

English

For each real number $\lambda \neq 7$, the system of equations

$$x + y + z = 3, \qquad x + 2y + 3z = 6, \qquad x + 4y + \lambda z = \lambda^2 - 7\lambda + 12$$

has a unique solution $(x_\lambda, y_\lambda, z_\lambda)$. Then $\lim\limits_{\lambda \to 7} y_\lambda$ is equal to

(A) 7
(B) -11
(C) 11
(D) -7

Only one option is correct.

हिन्दी

प्रत्येक वास्तविक संख्या $\lambda \neq 7$ के लिए समीकरण निकाय

$$x + y + z = 3, \qquad x + 2y + 3z = 6, \qquad x + 4y + \lambda z = \lambda^2 - 7\lambda + 12$$

का एक अद्वितीय हल $(x_\lambda, y_\lambda, z_\lambda)$ है। तब $\lim\limits_{\lambda \to 7} y_\lambda$ बराबर है

(A) 7
(B) -11
(C) 11
(D) -7

केवल एक विकल्प सही है।

Solution

Coefficient determinant: $\Delta = 1(2\lambda - 12) - 1(\lambda - 3) + 1(4 - 2) = \lambda - 7$, non-zero for $\lambda \neq 7$, so Cramer's rule applies.

Let $c = \lambda^2 - 7\lambda + 12$. Replacing the second column by the constants: $\Delta_2 = \begin{vmatrix}1 & 3 & 1\\ 1 & 6 & 3\\ 1 & c & \lambda\end{vmatrix} = 1(6\lambda - 3c) - 3(\lambda - 3) + 1(c - 6) = 3\lambda - 2c + 3.$

Substituting c: $\Delta_2 = 3\lambda - 2\lambda^2 + 14\lambda - 24 + 3 = -(2\lambda^2 - 17\lambda + 21) = -(2\lambda - 3)(\lambda - 7)$.

So $y_\lambda = \dfrac{\Delta_2}{\Delta} = -(2\lambda - 3) = 3 - 2\lambda$ for every $\lambda \neq 7$.

Similarly $\Delta_1 = \Delta_3 = c - 12 = \lambda(\lambda - 7)$, so $x_\lambda = z_\lambda = \lambda$; check: $\lambda + (3 - 2\lambda) + \lambda = 3$.

At $\lambda = 7$ itself $\Delta = 0$ and the system has infinitely many solutions (the constant becomes $12 = 3 \cdot 6 - 2 \cdot 3$); the limit picks out the solution $(7, -11, 7)$ of that line.

$\lim\limits_{\lambda \to 7} y_\lambda = 3 - 14 = -11$.

Correct Option: (B)

Question 07

English

Let p and q be integers with $1 \leq p \leq 10$ and $1 \leq q \leq 10$, and consider the system of equations

$$x + y + z = 3, \qquad x + 2y + pz = 4, \qquad x + 3y + 5z = q.$$

Then the number of ordered pairs (p, q) for which the system has a unique solution in which z is an integer is

(A) 19
(B) 90
(C) 12
(D) 21

Only one option is correct.

हिन्दी

माना कि p तथा q ऐसे पूर्णांक हैं कि $1 \leq p \leq 10$ तथा $1 \leq q \leq 10$, और समीकरण निकाय

$$x + y + z = 3, \qquad x + 2y + pz = 4, \qquad x + 3y + 5z = q$$

पर विचार कीजिए। तब उन क्रमित युग्मों (p, q) की संख्या, जिनके लिए निकाय का एक अद्वितीय हल है जिसमें z एक पूर्णांक है,

(A) 19
(B) 90
(C) 12
(D) 21

केवल एक विकल्प सही है।

Solution

Coefficient determinant: $\Delta = 1(10 - 3p) - 1(5 - p) + 1(3 - 2) = 6 - 2p$. The system has a unique solution exactly when $p \neq 3$; for $p = 3$ it has either none or infinitely many, so these pairs are excluded.

By Cramer's rule $z = \Delta_3/\Delta$ with $\Delta_3 = \begin{vmatrix} 1 & 1 & 3 \\ 1 & 2 & 4 \\ 1 & 3 & q \end{vmatrix} = 1(2q - 12) - 1(q - 4) + 3(3 - 2) = q - 5$.

So $z = \dfrac{q - 5}{6 - 2p}$, and z is an integer exactly when $6 - 2p$ divides $q - 5$, where $q - 5$ ranges over $-4, -3, \ldots, 5$.

$p = 2$ or $p = 4$: $6 - 2p = \pm 2$; the even values of $q - 5$ are $-4, -2, 0, 2, 4$, giving 5 pairs each, 10 in all.

$p = 1$ or $p = 5$: $6 - 2p = \pm 4$; the multiples of 4 are $-4, 0, 4$, giving 3 pairs each, 6 in all.

$p = 6, 7, 8, 9, 10$: $|6 - 2p| = 6, 8, 10, 12, 14$ exceeds $|q - 5|$ unless $q - 5 = 0$, so only $q = 5$ (where $z = 0$) works: 5 pairs.

Total $= 10 + 6 + 5 = 21$. Forgetting the pairs with $z = 0$ gives 12; counting all unique-solution pairs gives 90.

Correct Option: (D)

Question 08

English

For a complex number z, let

$$D(z) = \begin{vmatrix} 1 & z & z^2 \\ \bar{z} & 1 & z \\ \bar{z}^2 & \bar{z} & 7 \end{vmatrix},$$

where $\bar{z}$ denotes the complex conjugate of z. Then the minimum value of $D(z)$, as z varies over the set of all complex numbers, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

सम्मिश्र संख्या z के लिए माना कि

$$D(z) = \begin{vmatrix} 1 & z & z^2 \\ \bar{z} & 1 & z \\ \bar{z}^2 & \bar{z} & 7 \end{vmatrix}$$

है, जहाँ $\bar{z}$, z का सम्मिश्र संयुग्मी है। तब सभी सम्मिश्र संख्याओं z पर $D(z)$ का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $t = z\bar{z} = |z|^2 \geq 0$; then $\bar{z}z^2 = tz$, $\bar{z}^2 z = t\bar{z}$ and $\bar{z}^2 z^2 = t^2$.

Apply $R_2 \to R_2 - \bar{z}R_1$: the second row becomes $(0,\ 1 - \bar{z}z,\ z - \bar{z}z^2) = (0,\ 1 - t,\ z(1 - t))$.

Apply $R_3 \to R_3 - \bar{z}^2 R_1$: the third row becomes $(0,\ \bar{z} - \bar{z}^2 z,\ 7 - \bar{z}^2 z^2) = (0,\ \bar{z}(1 - t),\ 7 - t^2)$.

Expanding along the first column: $D = (1-t)(7-t^2) - z(1-t)\cdot\bar{z}(1-t) = (1-t)\left[7 - t^2 - t(1-t)\right] = (1-t)(7-t)$.

So $D(z)$ is real and depends only on $|z|$: $D = t^2 - 8t + 7$ with $t = |z|^2 \in [0, \infty)$.

$t^2 - 8t + 7 = (t-4)^2 - 9$, so $D \geq -9$, with equality when $t = 4$, i.e. on the circle $|z| = 2$ (for example $z = 2$: rows $(1, 2, 4)$, $(2, 1, 2)$, $(4, 2, 7)$, determinant -9).

Hence the minimum value is -9.

Final Answer: -9

Question 09

English

Let $f, g, h : \mathbb{R} \to \mathbb{R}$ be three times differentiable functions, each of which satisfies

$$\varphi'''(x) = \varphi''(x) + 4\varphi'(x) - 4\varphi(x) \quad \text{for all } x \in \mathbb{R},$$

with $f(0) = 1,\ f'(0) = 2,\ f''(0) = 0$; $g(0) = 0,\ g'(0) = 1,\ g''(0) = 3$; $h(0) = 2,\ h'(0) = 0,\ h''(0) = 1$. For $x \in \mathbb{R}$, let

$$F(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}.$$

Then the value of $F(\log_e 3)$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि $f, g, h : \mathbb{R} \to \mathbb{R}$ तीन बार अवकलनीय फलन हैं, जिनमें से प्रत्येक

$$\varphi'''(x) = \varphi''(x) + 4\varphi'(x) - 4\varphi(x), \quad x \in \mathbb{R}$$

को संतुष्ट करता है, तथा $f(0) = 1,\ f'(0) = 2,\ f''(0) = 0$; $g(0) = 0,\ g'(0) = 1,\ g''(0) = 3$; $h(0) = 2,\ h'(0) = 0,\ h''(0) = 1$ है। $x \in \mathbb{R}$ के लिए माना कि

$$F(x) = \begin{vmatrix} f(x) & g(x) & h(x) \\ f'(x) & g'(x) & h'(x) \\ f''(x) & g''(x) & h''(x) \end{vmatrix}$$

है। तब $F(\log_e 3)$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Differentiate the determinant row by row: $F'(x)$ is the sum of three determinants, the k-th having its k-th row differentiated.

Differentiating row 1 gives rows $(f', g', h'), (f', g', h'), (f'', g'', h'')$: two equal rows, value 0. Differentiating row 2 gives rows $(f, g, h), (f'', g'', h''), (f'', g'', h'')$: again 0.

So $F'(x) = \begin{vmatrix} f & g & h \\ f' & g' & h' \\ f''' & g''' & h''' \end{vmatrix}$.

Each function satisfies the given equation, so the third row is $(f'' + 4f' - 4f,\ g'' + 4g' - 4g,\ h'' + 4h' - 4h)$, which is $R_3 + 4R_2 - 4R_1$ in terms of the rows of F.

Subtracting $4R_2$ and adding $4R_1$ does not change a determinant, hence $F'(x) = F(x)$ for all x.

Then $\dfrac{d}{dx}\left(e^{-x}F(x)\right) = e^{-x}(F' - F) = 0$, so $F(x) = F(0)e^x$.

$$F(0) = \begin{vmatrix} 1 & 0 & 2 \\ 2 & 1 & 0 \\ 0 & 3 & 1 \end{vmatrix} = 1(1-0) - 0(2-0) + 2(6-0) = 13.$$

Therefore $F(\log_e 3) = 13e^{\log_e 3} = 39$. (The coefficients 4 and -4 never matter; only the coefficient 1 of φ'' decides the growth rate.)

Final Answer: 39

Question 10

English

Let a, b and c be the roots of the equation $2x^3 + 3x^2 - 5x + 1 = 0$. Then the value of

$$\frac{\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}}$$

is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a, b तथा c समीकरण $2x^3 + 3x^2 - 5x + 1 = 0$ के मूल हैं। तब

$$\frac{\begin{vmatrix} 1 & 1 & 1 \\ a^2 & b^2 & c^2 \\ a^3 & b^3 & c^3 \end{vmatrix}}{\begin{vmatrix} 1 & 1 & 1 \\ a & b & c \\ a^2 & b^2 & c^2 \end{vmatrix}}$$

का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Let $p(x) = 2x^3+3x^2-5x+1$. Then $p(-3) = -11 < 0$, $p(-2) = 7 > 0$, $p(0) = 1 > 0$, $p(\frac{1}{2}) = -\frac{1}{2} < 0$, $p(1) = 1 > 0$, so the three roots are real and distinct and the denominator is non-zero.

Denominator: the Vandermonde determinant $V = (b-a)(c-a)(c-b)$.

Numerator N: apply $C_2 \to C_2 - C_1$ and $C_3 \to C_3 - C_1$. The first row becomes $(1, 0, 0)$, so $N = \begin{vmatrix} b^2-a^2 & c^2-a^2 \\ b^3-a^3 & c^3-a^3 \end{vmatrix}$.

Take $(b-a)$ from the first column and $(c-a)$ from the second: $N = (b-a)(c-a)\begin{vmatrix} b+a & c+a \\ b^2+ab+a^2 & c^2+ca+a^2 \end{vmatrix}$.

The 2×2 determinant is $(b+a)(c^2+ca+a^2) - (c+a)(b^2+ab+a^2) = bc^2 - b^2c + ac^2 - ab^2 = bc(c-b) + a(c-b)(c+b) = (c-b)(ab+bc+ca)$.

Hence $\dfrac{N}{V} = ab+bc+ca$, and by Vieta's relations for $2x^3+3x^2-5x+1$: $ab+bc+ca = \dfrac{-5}{2} = -2.5$.

Final Answer: -2.50

Question 11

English

For real numbers x and λ, let

$$D(x,\lambda) = \begin{vmatrix} 1 & \cos x & \sin x \\ \cos x & 1 & \lambda \\ \sin x & \lambda & 1 \end{vmatrix}.$$

Then the maximum value of $D(x,\lambda)$, as x and λ vary over $\mathbb{R}$, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्याओं x तथा λ के लिए माना कि

$$D(x,\lambda) = \begin{vmatrix} 1 & \cos x & \sin x \\ \cos x & 1 & \lambda \\ \sin x & \lambda & 1 \end{vmatrix}$$

है। तब x तथा λ के $\mathbb{R}$ पर बदलने पर $D(x,\lambda)$ का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Expand along the first row: $D = 1\cdot(1-\lambda^2) - \cos x\,(\cos x - \lambda\sin x) + \sin x\,(\lambda\cos x - \sin x)$.
$= 1 - \lambda^2 - \cos^2 x + \lambda\sin x\cos x + \lambda\sin x\cos x - \sin^2 x$.
Using $\sin^2 x + \cos^2 x = 1$ and $2\sin x\cos x = \sin 2x$: $D(x,\lambda) = \lambda\sin 2x - \lambda^2$.
For a fixed λ, the largest value of $\lambda\sin 2x$ over x is $|\lambda|$ (take $\sin 2x = \pm 1$ with the sign of λ).
So $D \le |\lambda| - \lambda^2 = \frac{1}{4} - \left(|\lambda| - \frac{1}{2}\right)^2 \le \frac{1}{4}$.
Equality holds for $\lambda = \frac{1}{2}$ and $x = \frac{\pi}{4}$: $D = \frac{1}{2}\cdot 1 - \frac{1}{4} = \frac{1}{4}$.
Hence the maximum value is 0.25.

Final Answer: 0.25

Question 12

English

Let

$$D = \begin{vmatrix} 7 & \log_2 3 & \log_2 5 \\ \log_3 2 & 7 & \log_3 5 \\ \log_5 2 & \log_5 3 & 7 \end{vmatrix}.$$

Then the value of D is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि

$$D = \begin{vmatrix} 7 & \log_2 3 & \log_2 5 \\ \log_3 2 & 7 & \log_3 5 \\ \log_5 2 & \log_5 3 & 7 \end{vmatrix}$$

है। तब D का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By change of base, $\log_p q = \dfrac{\ln q}{\ln p}$, so the (i,j) entry for $i \neq j$ is $\dfrac{\ln p_j}{\ln p_i}$ with $(p_1, p_2, p_3) = (2, 3, 5)$.

Multiply row i by $\ln p_i$; the determinant is multiplied by $\ln 2 \cdot \ln 3 \cdot \ln 5$. Then divide column j by $\ln p_j$; the determinant is divided by the same product. So the value is unchanged.

After these operations the (i,j) entry is $\dfrac{\ln p_j}{\ln p_i} \cdot \dfrac{\ln p_i}{\ln p_j} = 1$ for $i \neq j$, and the diagonal entry is $7 \cdot \dfrac{\ln p_i}{\ln p_i} = 7$.

So $D = \begin{vmatrix} 7 & 1 & 1 \\ 1 & 7 & 1 \\ 1 & 1 & 7 \end{vmatrix}$.

Apply $C_1 \to C_1 + C_2 + C_3$ and take out 9: $D = 9\begin{vmatrix} 1 & 1 & 1 \\ 1 & 7 & 1 \\ 1 & 1 & 7 \end{vmatrix}$; then $R_2 - R_1$, $R_3 - R_1$ give $9\begin{vmatrix} 1 & 1 & 1 \\ 0 & 6 & 0 \\ 0 & 0 & 6 \end{vmatrix} = 9 \cdot 36 = 324$.

(With diagonal 1 instead of 7 the matrix would have rank one and D would be 0.)

Final Answer: 324

Question 13

English

For real numbers α, β and γ, let

$$D(\alpha, \beta, \gamma) = \begin{vmatrix} 1 & \sin(\alpha - \beta) & \sin(\alpha - \gamma) \\ \sin(\beta - \alpha) & 1 & \sin(\beta - \gamma) \\ \sin(\gamma - \alpha) & \sin(\gamma - \beta) & 1 \end{vmatrix}.$$

Then the maximum value of $D(\alpha, \beta, \gamma)$, as α, β and γ vary over $\mathbb{R}$, is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्याओं α, β तथा γ के लिए माना कि

$$D(\alpha, \beta, \gamma) = \begin{vmatrix} 1 & \sin(\alpha - \beta) & \sin(\alpha - \gamma) \\ \sin(\beta - \alpha) & 1 & \sin(\beta - \gamma) \\ \sin(\gamma - \alpha) & \sin(\gamma - \beta) & 1 \end{vmatrix}$$

है। तब α, β तथा γ के $\mathbb{R}$ पर बदलने पर $D(\alpha, \beta, \gamma)$ का अधिकतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Put $u = \alpha - \beta$, $v = \beta - \gamma$, $w = \gamma - \alpha$, so $u + v + w = 0$. By $\sin(-\theta) = -\sin\theta$: $\sin(\beta - \alpha) = -\sin u$, $\sin(\alpha - \gamma) = -\sin w$, $\sin(\gamma - \beta) = -\sin v$.

With $s_1 = \sin u$, $s_2 = \sin v$, $s_3 = \sin w$ the matrix is $I + S$ with S skew-symmetric: $D =$

$\begin{vmatrix} 1 & s_1 & -s_3 \\ -s_1 & 1 & s_2 \\ s_3 & -s_2 & 1 \end{vmatrix}$.

Expanding along the first row: $1(1+s_2^2) - s_1(-s_1 - s_2 s_3) + (-s_3)(s_1 s_2 - s_3) = 1 + s_2^2 + s_1^2 + s_1 s_2 s_3 - s_1 s_2 s_3 + s_3^2$.

So $D = 1 + \sin^2(\alpha - \beta) + \sin^2(\beta - \gamma) + \sin^2(\gamma - \alpha)$; the cubic terms cancel.

Compound angles: $\sin^2 u + \sin^2 v = 1 - \frac{\cos 2u + \cos 2v}{2} = 1 - \cos(u+v)\cos(u-v)$, and $\sin^2 w = \sin^2(u+v) = 1 - \cos^2(u+v)$.

With $c = \cos(u+v)$: $S = s_1^2 + s_2^2 + s_3^2 = 2 - c\cos(u-v) - c^2 \le 2 + |c| - c^2 = \frac{9}{4} - \left(|c| - \frac{1}{2}\right)^2 \le \frac{9}{4}$.

Equality needs $|c| = \frac{1}{2}$ and $c\cos(u-v) = -|c|$: take $u = v = \frac{2\pi}{3}$, so $u + v = \frac{4\pi}{3}$, $c = -\frac{1}{2}$, $\cos(u-v) = 1$; then each $\sin^2$ is $\frac{3}{4}$ and $S = \frac{9}{4}$.

Such angles exist, for example $\alpha = 0$, $\beta = -\frac{2\pi}{3}$, $\gamma = -\frac{4\pi}{3}$. Hence $D_{\max} = 1 + \frac{9}{4} = \frac{13}{4} = 3.25$.

Final Answer: 3.25

Question 14

English

Let $\vec{a}$, $\vec{b}$ and $\vec{c}$ be vectors such that $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 1$, $\vec{a} \cdot \vec{b} = 1$, $\vec{b} \cdot \vec{c} = 1$ and $\vec{c} \cdot \vec{a} = \frac{1}{2}$. Then the volume of the parallelepiped whose coterminous edges are $\vec{a}$, $\vec{b}$ and $\vec{c}$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि सदिश $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ इस प्रकार हैं कि $|\vec{a}| = 1$, $|\vec{b}| = 2$, $|\vec{c}| = 1$, $\vec{a} \cdot \vec{b} = 1$, $\vec{b} \cdot \vec{c} = 1$ तथा $\vec{c} \cdot \vec{a} = \frac{1}{2}$ है। तब उस समान्तर षट्फलक का आयतन, जिसकी संलग्न (coterminous) कोरें $\vec{a}$, $\vec{b}$ तथा $\vec{c}$ हैं, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Write $\vec{a} = (a_1, a_2, a_3)$ and so on, and let M be the matrix with rows $\vec{a}, \vec{b}, \vec{c}$. Then $\det M = [\vec{a}\ \vec{b}\ \vec{c}]$, the scalar triple product, whose absolute value is the volume.

Multiply $\det M$ by $\det M^T$ (equal to $\det M$) row by column: the (i,j) entry of the product is the dot product of the i-th and j-th vectors.

So $[\vec{a}\ \vec{b}\ \vec{c}]^2 = \begin{vmatrix} \vec{a}\cdot\vec{a} & \vec{a}\cdot\vec{b} & \vec{a}\cdot\vec{c} \\ \vec{b}\cdot\vec{a} & \vec{b}\cdot\vec{b} & \vec{b}\cdot\vec{c} \\ \vec{c}\cdot\vec{a} & \vec{c}\cdot\vec{b} & \vec{c}\cdot\vec{c} \end{vmatrix}$.

Substituting the data: $\begin{vmatrix} 1 & 1 & \frac{1}{2} \\ 1 & 4 & 1 \\ \frac{1}{2} & 1 & 1 \end{vmatrix} = 1(4-1) - 1\left(1 - \frac{1}{2}\right) + \frac{1}{2}(1-2) = 3 - \frac{1}{2} - \frac{1}{2} = 2$.

The data are consistent: the leading minors 1, $4 - 1 = 3$ and 2 are positive, so such vectors exist.

Hence the volume is $\sqrt{2} = 1.41421\ldots$, which is 1.41 to two decimal places.

Final Answer: 1.41

Question 15

English

For a real number t, let

$$D(t) = \begin{vmatrix} (1+t)^2 & (1+2t)^2 & (1+3t)^2 \\ (2+t)^2 & (2+2t)^2 & (2+3t)^2 \\ (3+t)^2 & (3+2t)^2 & (3+3t)^2 \end{vmatrix}.$$

Let t_0 be the real number for which $D(t_0) = 27$. Then the value of t_0 is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या t के लिए माना कि

$$D(t) = \begin{vmatrix} (1+t)^2 & (1+2t)^2 & (1+3t)^2 \\ (2+t)^2 & (2+2t)^2 & (2+3t)^2 \\ (3+t)^2 & (3+2t)^2 & (3+3t)^2 \end{vmatrix}$$

है। माना कि t_0 वह वास्तविक संख्या है जिसके लिए $D(t_0) = 27$ है। तब t_0 का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

The (i, j) entry is $(i + jt)^2 = 1 \cdot (j^2t^2) + i \cdot (2jt) + i^2 \cdot 1$ for $i, j \in \{1, 2, 3\}$.
So the matrix of $D(t)$ is AB, where A has rows $(1, i, i^2)$ for $i = 1, 2, 3$ and B has rows $(t^2, 4t^2, 9t^2)$, $(2t, 4t, 6t)$, $(1, 1, 1)$. By the multiplication theorem $D(t) = \det A \cdot \det B$.

$\det A = \begin{vmatrix} 1 & 1 & 1 \\ 1 & 2 & 4 \\ 1 & 3 & 9 \end{vmatrix}$ is a Vandermonde determinant: $(2-1)(3-1)(3-2) = 2$.

In $\det B$ take t^2 out of the first row and $2t$ out of the second: $\det B = 2t^3 \begin{vmatrix} 1 & 4 & 9 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix}$.

$\begin{vmatrix} 1 & 4 & 9 \\ 1 & 2 & 3 \\ 1 & 1 & 1 \end{vmatrix} = 1(2-3) - 4(1-3) + 9(1-2) = -1 + 8 - 9 = -2$, so $\det B = -4t^3$.

Hence $D(t) = -8t^3$ for all t (check: $D(1)$ has rows $(4, 9, 16), (9, 16, 25), (16, 25, 36)$ with determinant -8).

$-8t^3 = 27$ has the single real root $t_0 = -\frac{3}{2} = -1.5$.

Final Answer: -1.50

Question 16

English

For a real number λ, consider the system of equations

$$x + 2y - z = 0, \qquad 2x + \lambda y + 2z = 0, \qquad 3y + \lambda z = 0.$$

Let T be the set of all triples (x, y, z) of integers, not all zero, that satisfy this system for at least one real value of λ. Then the least value of $x^2 + y^2 + z^2$ for $(x, y, z) \in T$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या λ के लिए समीकरण निकाय

$$x + 2y - z = 0, \qquad 2x + \lambda y + 2z = 0, \qquad 3y + \lambda z = 0$$

पर विचार कीजिए। माना कि T पूर्णांकों के उन सभी त्रिकों (x, y, z), जिनमें सभी शून्य नहीं हैं, का समुच्चय है जो λ के कम से कम एक वास्तविक मान के लिए इस निकाय को संतुष्ट करते हैं। तब $(x, y, z) \in T$ के लिए $x^2 + y^2 + z^2$ का न्यूनतम मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A homogeneous system has a solution other than $(0, 0, 0)$ exactly when its coefficient determinant is zero.

$$\begin{vmatrix} 1 & 2 & -1 \\ 2 & \lambda & 2 \\ 0 & 3 & \lambda \end{vmatrix} = 1(\lambda^2 - 6) - 2(2\lambda - 0) + (-1)(6 - 0) = \lambda^2 - 4\lambda - 12 = (\lambda - 6)(\lambda + 2).$$

So only $\lambda = 6$ and $\lambda = -2$ give triples in T; for every other λ the only solution is the trivial one.

For these λ the first and third equations are independent (their coefficient rows are not proportional), so the solutions form a line, given by Cramer's rule for two equations in three unknowns: $x : y : z = \begin{vmatrix} 2 & -1 \\ 3 & \lambda \end{vmatrix} : -\begin{vmatrix} 1 & -1 \\ 0 & \lambda \end{vmatrix} : \begin{vmatrix} 1 & 2 \\ 0 & 3 \end{vmatrix} = (2\lambda + 3) : (-\lambda) : 3.$

$\lambda = 6$: $x : y : z = 15 : -6 : 3 = 5 : -2 : 1$. Check in equation 2: $10 - 12 + 2 = 0$. Integer solutions are $k(5, -2, 1)$ with k a non-zero integer, and $x^2 + y^2 + z^2 = 30k^2 \geq 30$.

$\lambda = -2$: $x : y : z = -1 : 2 : 3$. Check in equation 2: $-2 - 4 + 6 = 0$. Integer solutions are $k(-1, 2, 3)$, and $x^2 + y^2 + z^2 = 14k^2 \geq 14$.

Both directions are primitive (coordinates with no common factor), so no smaller integer multiples exist. The least value is 14, at $(-1, 2, 3)$ or $(1, -2, -3)$. Taking only the positive λ would give 30.

Final Answer: 14

Question 17

English

Let $R(x) = \alpha x^2 + \beta x + \gamma$ be the remainder when the polynomial x^6 is divided by $(x-1)(x-2)(x-3)$. Then the value of α is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि बहुपद x^6 को $(x - 1)(x - 2)(x - 3)$ से भाग देने पर शेषफल $R(x) = \alpha x^2 + \beta x + \gamma$ है। तब α का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

By the division algorithm $x^6 = (x - 1)(x - 2)(x - 3)Q(x) + R(x)$ with $\deg R \leq 2$.

Putting $x = 1, 2, 3$ (the remainder theorem for each linear factor) gives $R(1) = 1$, $R(2) = 2^6 = 64$, $R(3) = 3^6 = 729$.

So $\alpha + \beta + \gamma = 1$, $4\alpha + 2\beta + \gamma = 64$, $9\alpha + 3\beta + \gamma = 729$.

Coefficient determinant: $\Delta = \begin{vmatrix} 1 & 1 & 1 \\ 4 & 2 & 1 \\ 9 & 3 & 1 \end{vmatrix} = 1(2-3) - 1(4-9) + 1(12-18) = -1 + 5 - 6 = -2 \neq 0.$

Replace the first column by the constants: $\Delta_\alpha = \begin{vmatrix} 1 & 1 & 1 \\ 64 & 2 & 1 \\ 729 & 3 & 1 \end{vmatrix} = 1(2-3) - 1(64-729) + 1(192-1458) =$ $-1 + 665 - 1266 = -602.$

By Cramer's rule, $\alpha = \dfrac{\Delta_\alpha}{\Delta} = 301.$

Check: α is the second divided difference $\dfrac{729 - 2 \cdot 64 + 1}{2} = \dfrac{602}{2} = 301$; the full remainder is $301x^2 - 840x + 540$, and $R(2) = 1204 - 1680 + 540 = 64.$

Final Answer: 301

Question 18

English

For a real number k, consider the system of equations

$$kx + 2y - z = 1, \qquad 2x + (k+3)y - 2z = 1, \qquad -x - 2y + kz = 1.$$

Then the sum of all real values of k for which this system has no solution is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्या k के लिए समीकरण निकाय

$$kx + 2y - z = 1, \qquad 2x + (k+3)y - 2z = 1, \qquad -x - 2y + kz = 1$$

पर विचार कीजिए। तब k के उन सभी वास्तविक मानों का योग, जिनके लिए इस निकाय का कोई हल नहीं है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Expanding along the first row: $\Delta = k\,[k(k+3) - 4] - 2(2k-2) + (-1)(-4+k+3) = k^3 + 3k^2 - 4k - 4k + 4 - k + 1 = k^3 + 3k^2 - 9k + 5.$

$k^3 + 3k^2 - 9k + 5 = (k-1)^2(k+5)$. (Structure: the matrix is $(k-1)I + \mathbf{v}\mathbf{v}^T$ with $\mathbf{v} = (1, 2, -1)$, $|\mathbf{v}|^2 = 6$.)

For $k \neq 1, -5$, $\Delta \neq 0$ and Cramer's rule gives a unique solution, so these k are excluded.

$\Delta = 0$ is also exactly the condition for the associated homogeneous system (right-hand sides 0) to have non-trivial solutions; we now test the two values against the given constants.

$k = 1$: the equations are $x + 2y - z = 1$, $2(x + 2y - z) = 1$, $-(x + 2y - z) = 1$, that is $x + 2y - z = 1,\ \frac{1}{2},\ -1$. The three planes are parallel and pairwise disjoint, so there is no solution.

$k = -5$: the rank of the coefficient matrix is 2 (the minor $\begin{vmatrix} -5 & 2 \\ 2 & -2 \end{vmatrix} = 6 \neq 0$). Replacing the first column by the constants: $\Delta_1 = \begin{vmatrix} 1 & 2 & -1 \\ 1 & -2 & -2 \\ 1 & -2 & -5 \end{vmatrix} = 1(10-4) - 2(-5+2) - 1(-2+2) = 12 \neq 0.$

Since $\Delta = 0$ but $\Delta_1 \neq 0$, the system is inconsistent at $k = -5$ as well.

So the system has no solution exactly for $k = 1$ and $k = -5$, and the sum is -4.

Final Answer: -4

Question 19

English

Let p, q and r be real numbers such that no two of the planes

$$P_1: \ x + 2y + 2z = 1, \qquad P_2: \ 2x + py + 4z = 5, \qquad P_3: \ px + 8y + qz = r$$

have a common point, and the origin is equidistant from P_1 and P_3. Then the distance between the two of these planes that are nearest to each other is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि p, q तथा r ऐसी वास्तविक संख्याएँ हैं कि तलों

$$P_1: \ x + 2y + 2z = 1, \qquad P_2: \ 2x + py + 4z = 5, \qquad P_3: \ px + 8y + qz = r$$

में से किन्हीं दो का कोई उभयनिष्ठ बिन्दु नहीं है, तथा मूलबिन्दु P_1 तथा P_3 से समदूरस्थ है। तब इन तलों में से एक-दूसरे के निकटतम दो तलों के बीच की दूरी _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

If two planes have no common point they are parallel and distinct. So P_2 and P_3 are both parallel to P_1, whose normal is $(1, 2, 2)$.

P_2: $(2, p, 4)$ must be a multiple of $(1, 2, 2)$; the first and third components give the factor 2, so $p = 4$. Then P_2 is $x + 2y + 2z = \frac{5}{2}$, distinct from P_1.

P_3: $(p, 8, q) = (4, 8, q)$ must be $4(1, 2, 2)$, so $q = 8$, and P_3 is $x + 2y + 2z = \frac{r}{4}$.

The three planes are parallel, so the system of the three equations has no solution, and they are pairwise disjoint provided $\frac{r}{4} \neq 1$ and $\frac{r}{4} \neq \frac{5}{2}$.

Distance of the origin from $x + 2y + 2z = c$ is $\frac{|c|}{3}$. Equidistance from P_1 and P_3: $\frac{1}{3} = \frac{|r/4|}{3}$, so $\frac{r}{4} = \pm 1$. The value $\frac{r}{4} = 1$ makes P_3 coincide with P_1, which is excluded; hence $\frac{r}{4} = -1$, $r = -4$.

The planes are $x + 2y + 2z = c$ with $c = -1, \ 1, \ \frac{5}{2}$. Distances between parallel planes are $\frac{|c_1 - c_2|}{3}$: $\frac{2}{3}$ for $(-1, 1)$, $\frac{1}{2}$ for $(1, \frac{5}{2})$, $\frac{7}{6}$ for $(-1, \frac{5}{2})$.

The nearest pair is P_1 and P_2, at distance $\frac{1}{2} = 0.5$.

Final Answer: 0.50

Question 20

English

For real numbers λ, μ and ν, consider the system of equations

$$x + \lambda y + z = 1, \qquad 3x - 6y + \mu z = 2, \qquad -2x + 4y - 2z = \nu.$$

Let Δ be the determinant of its coefficient matrix, and let Δ_1, Δ_2, Δ_3 be the determinants obtained from Δ by replacing its first, second and third column, respectively, by the column of constants. Suppose that $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ and that the system has no solution. Then the value of $\lambda\mu$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

वास्तविक संख्याओं λ, μ तथा ν के लिए समीकरण निकाय

$$x + \lambda y + z = 1, \qquad 3x - 6y + \mu z = 2, \qquad -2x + 4y - 2z = \nu$$

पर विचार कीजिए। माना कि Δ इसके गुणांक आव्यूह का सारणिक है तथा Δ_1, Δ_2, Δ_3 वे सारणिक हैं जो Δ के क्रमशः पहले, दूसरे तथा तीसरे स्तम्भ को अचरों के स्तम्भ से बदलने पर प्राप्त होते हैं। माना कि $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ है तथा निकाय का कोई हल नहीं है। तब $\lambda\mu$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Cramer's rule decides the system only when $\Delta \neq 0$. When $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ the system may have infinitely many solutions or none.

Suppose the coefficient matrix has rank 2. Then two of its columns, say C_i and C_j, are linearly independent. The determinant Δ_k with the remaining column C_k replaced by the constant column B is $\det[\ldots C_i \ldots C_j \ldots B \ldots] = 0$, so B is a combination of C_i and C_j, and a solution exists (with the k-th unknown 0). This contradicts 'no solution'.

The rank is not 0, so it is 1: every row is a multiple of one row, i.e. the three planes are parallel.

The third row $(-2, 4, -2) = -2(1, -2, 1)$. Proportionality with the first row $(1, \lambda, 1)$ forces $\lambda = -2$.

The second row $(3, -6, \mu)$ must be $3(1, -2, 1)$, so $\mu = 3$.

Check: with $\lambda = -2$, $\mu = 3$ the equations are $x - 2y + z = 1$, $x - 2y + z = \frac{2}{3}$, $x - 2y + z = -\frac{\nu}{2}$; the first two planes are parallel and disjoint, so there is no solution for any ν. All columns of the coefficient matrix are multiples of $(1, 3, -2)^T$, so each Δ_i has two proportional columns and equals 0.

Other solutions of $\Delta = \Delta_1 = \Delta_2 = \Delta_3 = 0$ (such as $\lambda = -2$, $\nu = -2$ with $\mu \neq 3$) have rank 2 and infinitely many solutions, so they are excluded.

Hence $\lambda\mu = (-2)(3) = -6$.

Final Answer: -6

Question 21

English

Let a, b and c be real numbers such that the system of equations

$$x - 2y + 2z = 3, \qquad 2x + ay + bz = 1, \qquad 3x - 6y + cz = -12$$

has no solution and no two of the planes $P_1 : x - 2y + 2z = 3$, $P_2 : 2x + ay + bz = 1$, $P_3 : 3x - 6y + cz = -12$ have a common point. The line

$$L : \frac{x-1}{2} = \frac{y+1}{1} = \frac{z}{-1}$$

meets the plane, among P_1, P_2, P_3, that is farthest from the origin at the point Q. Then the z-coordinate of Q is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि a, b तथा c ऐसी वास्तविक संख्याएँ हैं कि समीकरण निकाय

$$x - 2y + 2z = 3, \qquad 2x + ay + bz = 1, \qquad 3x - 6y + cz = -12$$

का कोई हल नहीं है तथा तलों $P_1 :\ x - 2y + 2z = 3$, $P_2 :\ 2x + ay + bz = 1$, $P_3 :\ 3x - 6y + cz = -12$ में से किन्हीं दो का कोई उभयनिष्ठ बिन्दु नहीं है। रेखा

$$L :\ \frac{x-1}{2} = \frac{y+1}{1} = \frac{z}{-1}$$

तलों P_1, P_2, P_3 में से मूलबिन्दु से सबसे दूर स्थित तल को बिन्दु Q पर मिलती है। तब Q का z-निर्देशांक _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

Two planes without a common point are parallel and distinct, so the normals $(2, a, b)$ and $(3, -6, c)$ are multiples of $(1, -2, 2)$: $a = -4$, $b = 4$ (factor 2) and $c = 6$ (factor 3).

Dividing by these factors: $P_1 :\ x - 2y + 2z = 3$, $P_2 :\ x - 2y + 2z = \frac{1}{2}$, $P_3 :\ x - 2y + 2z = -4$. The constants are distinct, so the planes are pairwise disjoint and the system has no solution, as required.

The distance of the origin from $x - 2y + 2z = c$ is $\dfrac{|c|}{\sqrt{1+4+4}} = \dfrac{|c|}{3}$: 1 for P_1, $\frac{1}{6}$ for P_2, $\frac{4}{3}$ for P_3. So P_3 is the farthest.

A general point of L is $(1+2t,\ -1+t,\ -t)$. The direction $(2, 1, -1)$ has dot product $2-2-2 = -2 \neq 0$ with the normal, so L is not parallel to the planes and meets each in exactly one point.

On L: $x - 2y + 2z = (1 + 2t) - 2(-1 + t) + 2(-t) = 3 - 2t$.

For P_3: $3 - 2t = -4$, so $t = \frac{7}{2}$ and $Q = \left(1 + 7,\ -1 + \frac{7}{2},\ -\frac{7}{2}\right) = \left(8,\ \frac{5}{2},\ -\frac{7}{2}\right)$. Check: $8 - 5 - 7 = -4$.

Hence the z-coordinate of Q is $-\frac{7}{2} = -3.5$.

Final Answer: -3.50

Question 22

English

Let k be the real number for which the system of equations

$$2x + y - z = 0, \qquad x - 3y + 2z = 0, \qquad kx + y - z = 0$$

has a non-trivial solution. For this value of k, the number of solutions (x, y, z) of the system in which x, y and z are integers with $|x| \leq 1000$, $|y| \leq 1000$ and $|z| \leq 1000$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि k वह वास्तविक संख्या है जिसके लिए समीकरण निकाय

$$2x + y - z = 0, \qquad x - 3y + 2z = 0, \qquad kx + y - z = 0$$

का एक अतुच्छ (non-trivial) हल है। k के इस मान के लिए निकाय के उन हलों (x, y, z) की संख्या, जिनमें x, y तथा z पूर्णांक हैं तथा $|x| \leq 1000$, $|y| \leq 1000$, $|z| \leq 1000$ है, _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A homogeneous system has a non-trivial solution exactly when its coefficient determinant is zero.

$\begin{vmatrix} 2 & 1 & -1 \\ 1 & -3 & 2 \\ k & 1 & -1 \end{vmatrix} = 2(3-2) - 1(-1-2k) + (-1)(1+3k) = 2 + 1 + 2k - 1 - 3k = 2 - k$, so $k = 2$.

For $k = 2$ the third equation is the same as the first, and the first two are independent, so the solutions form a line through the origin.

Cross-multiplying the first two equations: $x : y : z = (2-3) : (-1-4) : (-6-1) = -1 : -5 : -7 = 1 : 5 : 7$. So the solutions are $(t, 5t, 7t)$, $t \in \mathbb{R}$.

Since $x = t$, the solution is an integer triple exactly when t is an integer.

The bounds: $|t| \le 1000$, $|5t| \le 1000$, $|7t| \le 1000$; the last is the strongest, $|t| \le \frac{1000}{7} = 142.8\ldots$, so $t = -142, \ldots, 142$.

That is $2 \cdot 142 + 1 = 285$ solutions, the trivial solution $(0, 0, 0)$ included.

Final Answer: 285

Question 23

English

Let S be the set of all $\theta \in (0, \pi)$ for which the system of equations

$$x \sin\theta + y \cos\theta + z = 0, \qquad x \cos\theta + y \sin\theta + z = 0, \qquad x + y + 2z = 0$$

has a non-trivial solution. Then the value of $\dfrac{1}{\pi} \displaystyle\sum_{\theta \in S} \theta$ is _____.

The answer is a numerical value, correct up to the 2nd decimal digit.

हिन्दी

माना कि S, $(0, \pi)$ में उन सभी θ का समुच्चय है जिनके लिए समीकरण निकाय

$$x \sin\theta + y \cos\theta + z = 0, \qquad x \cos\theta + y \sin\theta + z = 0, \qquad x + y + 2z = 0$$

का एक अतुच्छ (non-trivial) हल है। तब $\dfrac{1}{\pi} \displaystyle\sum_{\theta \in S} \theta$ का मान _____ है।

उत्तर एक संख्यात्मक मान है (दशमलव के दूसरे अंक तक सही)।

Solution

A homogeneous system has a non-trivial solution exactly when its coefficient determinant is zero.

Expanding along the first row: $\sin\theta(2\sin\theta - 1) - \cos\theta(2\cos\theta - 1) + 1(\cos\theta - \sin\theta) = 2\sin^2\theta - 2\cos^2\theta + 2\cos\theta - 2\sin\theta$.

$= 2(\sin\theta - \cos\theta)(\sin\theta + \cos\theta) - 2(\sin\theta - \cos\theta) = 2(\sin\theta - \cos\theta)(\sin\theta + \cos\theta - 1)$.

First factor: $\sin\theta = \cos\theta \iff \tan\theta = 1 \iff \theta = n\pi + \frac{\pi}{4}$, $n \in \mathbb{Z}$. In $(0, \pi)$: $\theta = \frac{\pi}{4}$.

Second factor: $\sin\theta + \cos\theta = 1 \iff \sqrt{2}\sin\left(\theta + \frac{\pi}{4}\right) = 1 \iff \sin\left(\theta + \frac{\pi}{4}\right) = \sin\frac{\pi}{4}$.

General solution: $\theta + \frac{\pi}{4} = n\pi + (-1)^n \frac{\pi}{4}$, i.e. $\theta = 2m\pi$ (even n) or $\theta = (2m+1)\pi - \frac{\pi}{2} = 2m\pi + \frac{\pi}{2}$ (odd n). In the open interval $(0, \pi)$ only $\theta = \frac{\pi}{2}$; $\theta = 0$ is excluded.

Check $\theta = \frac{\pi}{2}$: the equations become $x + z = 0$, $y + z = 0$, $x + y + 2z = 0$, satisfied by $(1, 1, -1)$.

So $S = \left\{\frac{\pi}{4}, \frac{\pi}{2}\right\}$ and $\dfrac{1}{\pi}\left(\frac{\pi}{4} + \frac{\pi}{2}\right) = \frac{3}{4} = 0.75$.

Final Answer: 0.75